

Literature Citation Verification System Based on DeepScientist

Project A — Engineering Track: Evidence Chain Module

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STA304 Final Project · LLMs: From Principles to Applications

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The Problem

Core Pain Point

Research Agents (DeepScientist, OpenClaw) generate **professional-looking reports**, but users cannot answer three questions:

- ❶ **Which conclusions come from real sources?** — citations untraceable
- ❷ **Which are model inferences?** — no hallucination detection
- ❸ **Which have no evidence at all?** — report wording stronger than evidence

Our Goal

Build an **auditable, traceable evidence-chain system** that verifies every claim:

Upload PDF → Structured Evidence ID → API Verification → Scoring → Auto Audit

System Architecture

Pipeline Flow

Row 1 (Extraction)

QQ User $\longrightarrow$ crossdisc_idea (C) $\longrightarrow$ parse_pdf (A) $\longrightarrow$
verify_claim $\times N$ (A)

Row 2 (Scoring & Output)

score_batch (B) $\longrightarrow$ render_report (B) $\longrightarrow$ artifact_record +
memory_write (C) $\longrightarrow$ QQ Returns RYG Report

Three Roles

Person A — Tool Layer claim_extractor, semantic_verifier (PyPDF2,
S2/arXiv/Crossref API), HARD STOP RULE in builder.py

Person B — Scoring Layer traffic_light (7-branch RYG), render (Markdown),
MCP 5 tool registration, 6 Quest execution

FactCheck Pipeline Detail

Phase 0 — PDF Parsing

PyPDF2 extraction → regex detects
“REFERENCES” heading → split by [N] /
[Author,Year] → Unicode quote matching →
truncation (200K chars, 40 claims max)

Phase 1 — Claim Verification

3-tier fallback: (1) Semantic Scholar, (2) arXiv
Title API, (3) Crossref DOI search. Output:
verdict + confidence + evidence snippet.

Phase 2 — Batch Scoring (B)

supported, $\text{conf} \geq 0.7$	● Green
supported, $\text{conf} < 0.7$	● Yellow
not_found, uncertain	● Yellow
contradicted, $\text{conf} \geq 0.7$	● Red
contradicted, $\text{conf} < 0.7$	● Yellow

Phase 3 — Render (B)

Rendered Markdown with: summary table (RYG
counts), per-claim detail cards (verdict +
confidence + evidence snippet), Evidence Chain
table (E00X → C00X mapping).

Phase 4 — Persist (C)

artifact__record → evidence_store
memory__write → “episodes” kind.

Data Flow

Claim (C001) → VerificationResult →
ScoredClaimResult → FactCheckResult

Enforcement — Dual-Layer

Layer 1 (SKILL.md): 5-State Machine

3 Evidence Entry Types

Type	ID	Metadata
QQ Text	E001	text + SHA256
QQ Image	E001-img	dimensions + SHA256
QQ PDF	E001-pdf	pages + SHA256 + full-text

Final Report Structure

- §1 FactCheck Results (rendered RYG table)
- §2 Evidence Chain Table (E00X ↔ C00X, 40 rows)
- §3–5 Cross-discipline bridges + hypothesis
- §6 Caveats (verifier limits, hedging)

Audit: `audit_report()`

- Scans report for [E001], [E001-img], [E001-pdf] patterns
- Cross-references every cited ID against `evidence_store.json`
- Detects: fake IDs, unused evidence, bare claims (no annotation)
- Counts: supported / inferred / unverified claims
- Outputs PASS or WARN verdict

Three-Tier Annotation

[E001] Evidence-supported claim (● green)

Inferred Logically inferred, not directly proven

Needs Verification Insufficient evidence, needs further check

Test Results: 6 Disciplines × 6 Quests

Quest	Discipline	Claims	verify	Score	Tools
027	ML/Federated	40	85	G17 Y18 R5	✓
028	Medical Edu.	8	27	G1 Y6 R1	✓
029	Chemistry	1	7	G1 Y0 R0	✓
030	Optimization	40	105	G10 Y28 R2	✓
031	Physics	40	94	G2 Y37 R1	✓
032	ML Survey	1	5	G0 Y1 R0	✓

Key Metrics

139 tests all passing | **996 events** | **323 verify_claim** calls | **36/36** mandatory tools

Failure Case Analysis

- ❶ **Verifier False Positive (LIGO Q031)** C036 (GW190707) flagged • contradicted. Cause: S2 matched different paper with overlapping authors. Handling: Caveats states “almost certainly a false negative.”
- ❷ **Abstract-Only Limitation (SGD Q030)** C018/C031 unverifiable — theorems need full-text access, abstracts lack detail.
- ❸ **PDF Parser Fallback (Medical Q028)** `parse_pdf` returned 0 claims (PubMed-indexed, not arXiv). Agent auto-fallback to `bash_exec`. Extraction method marked as `bash_fallback` in evidence table.
- ❹ **Agent Bypassed Toolchain (mid-dev, fixed)** Early runs: 10/40 `verify_call`, 0 `score_batch`, 0 `render_report`. Fix (3 iterations): MUST → 5-State Machine → HARD STOP RULE. Result: 36/36 compliance.

Boundary Cases

Case	Quest	Description
Minimal claims	Q029	Only 1 claim extracted; system correctly outputs PASS
Max uncertainty	Q031	37/40 ●; system honestly explains abstract-only limit
Cross-discipline refs	Q031	LIGO astro-ph format not recognized by S2; marked not_found
Internal proofs	Q027/030	Theorems/lemmas have no external citation target; correctly flagged
Multi-citation	Q027	[20]--[24] dash-range parsed into individual markers
Fake ID detection	test	E999-img caught by audit_report as non-existent

Before / After Comparison

Dimension	Before (Original)	After (Our System)
Message Recording	QQ logs, no structure	E00X evidence_store (16 fields)
Citation Annotation	None	[E001] + RYG tri-color labels
Reference Verification	None	S2 / arXiv / Crossref API
Image Evidence	No metadata	SHA256 + dims + format
PDF Evidence	Not supported	Pages + SHA256 + full-text sidecar
Audit Capability	Impossible	audit_report auto fake-ID check
Experiment Logging	None	memory_write structured schema
Test Coverage	0	139 tests / 100% pass
Grading Level	C+ (prompt only)	≥ B- (MCP toolchain)

Current Limitations

- Abstract-only verification: most ● resolvable with full-text
- Verifier false positives: ~5–10% ● may be misclassifications
- PyPDF2 weak on scanned PDFs
- API-dependent: offline mode not supported
- No video/audio support yet

Future Improvements

- Full-text verification via arXiv open-access
- Local model fallback (Ollama + sentence-transformers)
- Structured NLP pipeline for claim extraction
- Multi-connector support (WeChat, Discord)
- Caching layer to avoid duplicate API calls

Team Contributions

Member	Role	Core Contribution
Miaoyi Liao	Person A (Tool)	claim_extractor, semantic_verifier, HARD STOP RULE, evidence-chain store
Xiaoyu Xiong	Person B (Scoring)	traffic_light (7 branches), render (Mark-down), MCP 5-tool registration, 6 Quest runs
Yuxiang Liu	Person C (Integration)	crossdisc_idea 5-Phase, QQ multi-modal evidence, daemon integration, memory schema

Cross-Collaboration Highlights

Phase 5 Enforcement: A (HARD STOP RULE) + B (5-State Machine) + C (memory/artifact schema) — three layers guarantee agent cannot bypass toolchain

Evidence Chain: A (storage + runner hooks) + B (audit + prompt protocol) + C (QQ multi-modal entry point) — full-link coverage

From “LLM Writes a Report” to “Agent Calls a Verifiable Toolchain”

Input	Verification	Output
6 disciplines	323 API lookups	RYG colored report
QQ real interaction	3-source fallback	Evidence Chain table
Text + Image + PDF	139 tests / 100%	Cross-disc. research ideas

Grade: $\geq$ **B-** (independent MCP toolchain, 36/36 mandatory calls)

https://github.com/leoplasture/STA304_Final_Project